LAB 3 / WIRING DIAGRAM

STM32F746G-DISCO + 1602A LCD + 3 LEDs + 2 buttons

Solid: code-defined signals. Dashed: inferred hookup or unconfirmed details.

1. LCD - 4-bit parallel interface

BOARD HEADER / MCU		LCD PIN / SIGNAL
A5 / PF6	———	4 RS
A4 / PF7	———	6 E
A0 / PA0	———	11 DB4
A1 / PF10	———	12 DB5
A2 / PF9	———	13 DB6
A3 / PF8	———	14 DB7
+5 V (inferred)	- - - - - - - - - -	2 VDD
GND (common)	- - - - - - - - - -	1 VSS, 5 R/W, 16 K
Contrast tap (section 3)	- - - - - - - - - -	3 VO
Backlight supply + limiting	- - - - - - - - - -	15 A / LED+

LCD pins 7-10 (DB0-DB3): unused. Preserve the working backlight current limiting.

2. LEDs and buttons

330 ohm each

D7 / PI3 - - [ ] - - green LED -> GND

D3 / PB4 - - [ ] - - orange LED -> GND

D4 / PG7 - - [ ] - - blue LED -> GND

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D13 / PI1 ---- SW1 ---- GND
D8  / PI2 ---- SW2 ---- GND
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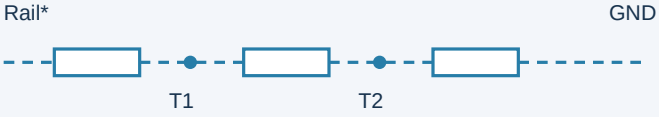
Buttons: internal pull-ups; press connects to GND.

3. VO contrast divider

Three 330 ohm resistors in series = 990 ohm



LCD VO -> T1 OR T2: exact tap not recorded.

*Actual rail voltage/end connections unconfirmed.

Illustration only; neither tap is claimed as verified.

READ BEFORE REWIRING

LED paths are inferred: GPIO -> resistor -> anode; cathode -> common GND.

A5/A4 must route to PF6/PF7, rather than their alternate I2C connections.

LCD supply and backlight hookup are inferred; confirm the actual module labels and working setup.

RESULT: All three LEDs, TIM2 sequence, both interrupts, and LCD reported working.

Installing the contrast resistor chain resolved the LCD readability issue. VO was not measured.

Pin references: ST UM1907 Arduino connector table; standard 16-pin LCD1602 interface.

GPIO mapping checked against supplied lab code. Physical wiring was not independently inspected.